

# ECON 311

## Lecture 9: Economics of Ideas

# Introduction

---

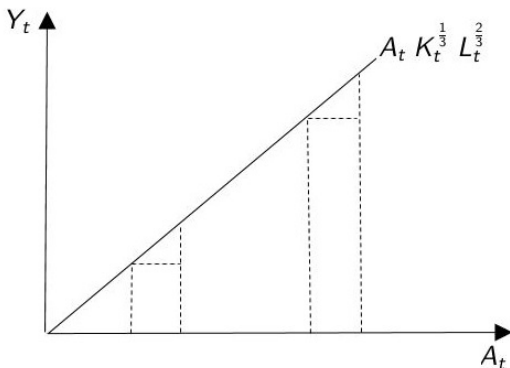
- A large share of differences in  $y$  across countries is due to differences in  $\bar{A}$ 
  - ▶ Differences in  $\bar{A}$  are particularly important in understanding differences between rich and poor countries
  - ▶ Any policy aimed to reduce the gap between rich and poor countries needs to target the large differences in  $\bar{A}$
  - ▶ To understand differences in  $\bar{A}$ , we need a model that does not treat TFP as an exogenous parameter
- The Romer Model incorporates differences in TFP by directly modeling the economics of ideas.
  - ▶ Ideas are a different kind of input of production
  - ▶ The production of ideas is not free

## Ideas in the Production Function

- Production of ideas result in changes in TFP:

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{\frac{1}{3}} L_t^{\frac{2}{3}}$$

- There are constant returns to TFP



## Ideas in the Production Function

- Production of ideas result in changes in TFP:

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{\frac{1}{3}} L_t^{\frac{2}{3}}$$

- There are increasing returns to scale

- ▶ Multiply all inputs by a constant ( $\gamma$ ):

$$Y_t = F(\gamma K_t, \gamma L_t, \gamma A_t) = \gamma A_t (\gamma K_t)^{\frac{1}{3}} (\gamma L_t)^{\frac{2}{3}}$$

$$Y_t = \gamma^2 F(K_t, L_t, A_t)$$

# The Production of Ideas

---

- Producing ideas is not free:
  - ▶ Innovators must invest “labour” to produce ideas:

$$L_{A_t} = \bar{l} \bar{L}$$

$$L_{Y_t} = (1 - \bar{l}) \bar{L}$$

$$L_{A_t} + L_{Y_t} = \bar{L}$$

- ▶ Trade-off between producing consumption goods and ideas:

$$Y_t = A_t L_{Y_t}$$

$$A_{t+1} = A_t + \underbrace{\bar{z} A_t L_{A_t}}_{\text{Idea}}$$

# TFP Growth

- Accumulation of ideas result in TFP growth over the long-run:

$$\underbrace{\frac{A_{t+1} - A_t}{A_t}}_{\bar{g}} = \bar{z} \bar{l} \bar{L}$$

- (Trade-off) An increase in  $\bar{l}$  increases  $A_{t+1}$ , but decreases  $Y_t$ 
  - ▶ An increase in  $\bar{l}$  has the potential to increase output in the long-run
  - ▶ Furthermore, an increase in  $\bar{l}$  increases the long-run growth rate of output

# Output Growth in the Long-Run

- Output per capite at time  $t$  is:

$$y_t = A_0 (1 - \bar{l}) (1 + \bar{g})^t$$

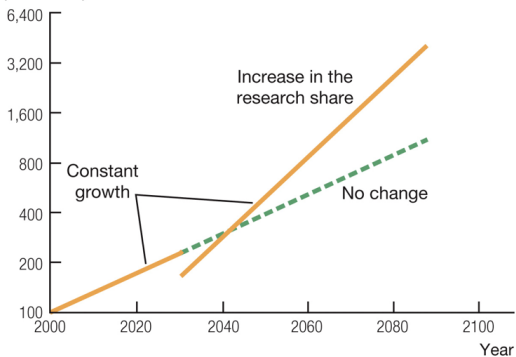
- Using the growth rate of TFP, we can solve for the long-run growth of  $y_t$  (BGP):

$$g_y = \bar{z} \bar{l} \bar{L} (1 - \bar{l})$$

# Trade-Off

Output per Person after an Increase in  $\bar{\ell}$

Output per person,  $y_t$   
(ratio scale)



# Conclusion

---

- Introducing the production of ideas ...
  - ▶ provides a theory as to why, in the long-run, some countries are able to grow at a faster rate than others
    1. More labour allocated to the ideas producing sector
    2. Countries have different ideas generating processes
  - ▶ is consistent with the data because it models long-run growth in GDP per capita as the result of TFP growth